

CAD Domain

Task 1 — Nano + MPU6050 Enclosure with Custom Lid

Components:

- Arduino Nano — mechanical drawing attached (see [Nano_drawing](#)). 4× Ø1.8mm mounting holes.
- MPU6050 breakout — mechanical drawing attached (see [MPU6050_drawing](#)). 4× Ø2.5mm mounting holes.

Requirement: Design an enclosure with a lid that houses both boards. The enclosure must:

- Mount both PCBs internally on standoffs, using the exact hole patterns from the attached drawings — boards must not sit flush against the base.
- Include a perimeter wall around the base. This is mandatory — a flat baseplate with standoffs and no surrounding wall does not satisfy the task.
- Include a lid that closes over the full footprint. The lid mechanism (screws, snap-fit, slide-fit, or any other approach) is entirely your choice — this is an area of creative freedom, but the lid must be removable/openable to access the internals for maintenance.
- Keep the Nano's USB port accessible from outside the enclosure without needing to remove the lid or either board.
- Account for the Nano's onboard LEDs remaining visible or otherwise accessible despite the lid (a window, cutout, or any other approach of your choosing).
- Maximum external footprint: 70mm (L) × 50mm (W) × 25mm (H), lid included (i.e. this is the fully closed, assembled size). Both boards must fit within this envelope in any orientation you choose — how you arrange the two boards inside this space is part of the design problem.

Deliverable:

- 3D model file (STL or STEP)
- A short design rationale (150–250 words): board arrangement, standoff heights, wall height, your lid mechanism choice and why, and any tradeoffs made to fit the envelope.

Task 2 — Slider-Crank Mechanism

Requirement: Design a functional slider-crank mechanism: a rotating crank link connects via a connecting rod to a slider constrained to move in a straight line, converting continuous rotary input into reciprocating linear output.

The assembly must be properly constrained using real joints/mates in your CAD software (revolute joints at the crank pivot and crank–rod junction, a revolute joint at the rod–slider junction, and a prismatic/slider constraint restricting the slider to a single line of travel) — not bodies that are merely visually positioned to look correct.

Rough sizing guidance:

- Crank radius: roughly 15–30mm.
- Connecting rod length: roughly 2–4× the crank radius.
- Expect a slider travel/stroke of roughly 2× the crank radius.
- You may deviate from these values, but justify your choice if you do.

Deliverable:

- Assembly file (STEP preferred, to preserve joint/mate definitions)
 - A short video/gif showing the mechanism at several crank angles (e.g. 0°, 90°, 180°, 270°) demonstrating it moves through its full range without interference or binding
 - A design rationale (150–250 words): your crank radius/rod length choice and why, how the joints were defined, and any interference issues you checked for.
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